

Part A – Conceptual Foundation (Theory)

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1. What is Regularization in Machine Learning? Why is it needed?

Regularization is a machine learning technique used to reduce overfitting by adding a penalty term to the model's loss function. It discourages the model from learning overly complex relationships and helps improve its performance on unseen data. Regularization improves model generalization and makes predictions more stable.

2. Difference Between Ridge Regression (L2) and Lasso Regression (L1)

Ridge Regression (L2)

Uses L2 regularization.

Shrinks coefficients but rarely makes them zero.

Keeps all features in the model.

Best when most features are useful.

More stable with correlated features.

Lasso Regression (L1)

Uses L1 regularization.

Can shrink coefficients to exactly zero.

Performs automatic feature selection.

Best when only a few features are important.

Produces simpler and more interpretable models.

3. What is Cross-Validation and Why is it Important?

Cross-validation is a model evaluation technique used to estimate how well a machine learning model will perform on unseen data. It divides the dataset into multiple training and validation sets, ensuring that every observation is used for both training and testing. It helps reduce overfitting and provides a more reliable estimate of model performance.

4. Cross-Validation Techniques

a) K-Fold Cross-Validation

K-Fold Cross-Validation divides the dataset into **K equal parts (folds)**. The model is trained on K-1 folds and tested on the remaining fold. This process repeats until every fold has been used as the validation set, and the average performance is calculated.

b) Stratified K-Fold Cross-Validation

Stratified K-Fold ensures that each fold has approximately the same distribution of the target variable as the original dataset. It is commonly used for classification problems and helps reduce bias caused by imbalanced data.

c) Leave-One-Out Cross-Validation (LOOCV)

LOOCV is a special case of K-Fold Cross-Validation where the number of folds equals the number of observations. In each iteration, one sample is used for testing while the remaining samples are used for training. It provides a nearly unbiased estimate but is computationally expensive.

d) Time Series Split

Time Series Split is used for time-dependent data. Instead of randomly splitting the dataset, it trains the model on past observations and validates it on future observations. This preserves the chronological order of the data and avoids data leakage.

5. Why are Tree-Based Models Less Sensitive to Feature Scaling?

Tree-based models such as Decision Trees and Random Forests split data based on feature values rather than distance calculations. Since they rely on threshold-based decisions instead of measuring distances, feature scaling has little or no effect on their performance. Therefore, normalization or standardization is generally not required for tree-based algorithms.